

EUNICO CORNELIUS

Associate Software Engineer (spc. AI) | Melbourne VIC

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I am an Associate Software Engineer with a Masters of Software Engineering (spec. AI) from The University of Melbourne with 11+ years of coding experience using technologies including Python, Javascript and Java. Recently, I focused on Full-stack and Artificial Intelligence system development to tackle real industry problems. My front-end tech-stack includes, React, TS, and Tailwind (CSS); backend development using Python (FastAPI) with MongoDB or Postgres SQL; deployment using Vercel, Cloudflare and GH-Pages; version control using Git. For machine learning development, I use Python with SciKit library, Numpy, Pandas, Matplotlib both for supervised and unsupervised learning along with Tensorflow + Keras for tensor based learning. Additionally, I am also excellent in prompt engineering using claude code for improved development efficiency and debugging.

SKILLS

- **Programming:** Python, Javascript, Java, HTML/CSS, ReactJS, Redux, React Native, PHP, C#, ASP.net, VueJS, Quasar, Flutter, MongoDB, XML, Go.
- **Software/Libraries:** Git, Bitbucket, Unity, Docker, Processwire, Android Studio, Xcode, Adobe Photoshop, Microsoft Office.
- **Packages:** SQLAlchemy, FastAPI, Tensorflow, pandas, Keras, matplotlib, opencv, octave, scikit-learn, tesseract, Flask, Jquery, Cypress, D3JS, Werkzeug, Bcrypt, FlaskJWT, Minio.
- **Learning:** Cloudflare, AWS services, Kubernetes, ThreeJS, QGIS/ArcGIS, SQLite.

ACHIEVEMENTS

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| • SecDim Secure Coding Competition Top 3 | 2026 |
| • Panel speaker for University of Melbourne's MDHS faculty conference | 2024 |
| • Airwallex Excellence in Technology Masters Scholarship | 2023 |
| • Jack Wynhoven Scholarship | 2023 |
| • Winner of 2023 IT Hackathon Festival "Most Aesthetic Project" | 2023 |
| • 1st Place in Digital Transformation of Health Hackathon | 2023 |

PROFESSIONAL EXPERIENCE

AURIN - Urban & Infrastructure Data Company

Associate Software Engineer

March 2025 – Current

- Develop and maintain 4.822 dataset and API gateways for 3+ major Australian data providers such as DSpark, Domain and Nearmap, enabling seamless access to high-value datasets for public and private research analytics.
- Increase quality of shared multi-container UI/UX components and authentication modules across multiple services (including CKAN, Frontend, and Backend) to establish a unified and maintainable application framework.
- Implement secure authentication and authorization mechanisms using AAF, OIDC, and Keycloak, supporting trusted access to sensitive and agreement-oriented datasets.
- Manage repository dependencies, internal tools, and automated github workflows, ensuring consistent testing, deployment reliability, and efficient collaboration within the engineering team across 10+ containerised docker services.
- Ensure robust code quality and maintainability through code reviews, automated cypress testing, and continuous integration practices that uphold high software reliability standards.
- Contribute to a culture of technical excellence and collaboration through active upskilling, mentoring, and cross-team engagement to create a positively impactful digital infrastructure.
- Present and facilitate in multiple company meetings to provide clear technical outcomes, demonstrations, and progress fostering transparency and knowledge sharing across the organisation.
- Advocate for the needs of Australian researchers and data users through empathetic design thinking, user journey mapping by utilising prototyping skills (Figma) that provides a solution to any problem within a week or less.
- Establish and maintain SEO performance standards using lighthouse and locust to ensure Google index searchability.
- Implement early-stage prototyping for technological advancements in AI including MCPs (Claude), Langchain and NVIDIA NIM.

Casual Software Engineer

September 2024 – March 2025

- Develop and enhance applications and APIs using Python, Java, JavaScript (React), and Go, integrating production-ready data services with React to support scalable digital infrastructure for research and innovation across Australia.

- Maintain and optimize automated CI/CD testing pipelines in both cloud and local environments, ensuring stable and efficient deployments through well-structured Docker containers and GitHub-based workflows.
- Collaborate in an Agile environment, multidisciplinary team to improve monorepo systems and strengthen collaborative development practices that underpin reliable national data delivery.
- Design and implement Grafana dashboards to visualise performance metrics, perform diagnostic and forensic analysis on logs and improve transparency in service operations to detect operational anomalies and identify potential security abnormalities.
- Prepare and maintain technical documentation and deployment procedures for both internal and external stakeholders through Github, Confluence and tutorials.
- Provide technical input into project planning, design and resource estimation for future software initiatives through Miro and Trello.

Hanson Australia - Construction Company

November 2023 – February 2024

Web Developer

- Published a new system with 5 fully functional and responsive pages using Vite, React, and .NET stack.
- Established 3 reusable and scalable react components using Typescript and Syncfusion.
- Maintain a high level of communication directly with the client in continuous development.

Labster ApS - Educational Software Company

January 2021 – September 2022

Content Developer

- Boosted production efficiency by 300% using scalable design patterns and agile methods using Jira (SCRUM and Kanban).
- Built reusable code templates and building blocks that are reusable developers to quickly implement for multiple other simulations.
- Developed, managed and released into production more than 30 higher education simulations with topics in natural sciences including stereochemistry, synthesis of aspirin, melting point apparatus and more with accessibility friendly features via XML and Unity.
- Accessibility supported, version controlled, and maintained on 20+ simulations that satisfy quality assured standards for production.

Cognixy, Jakarta - Software Start-Up

June – September 2020

Full Stack Developer

- Developed an Intelligent Attendance System that utilizes neural networks to detect human beings and their thermal signatures to prevent covid spread with Python (Flask), ZMQ, Redis, MySQL, Quasar and VueJS.
- Built attendance mailing system that envelops NN model above to notify employers of COVID symptomatic employees.
- Product bought by ShapesAI and used for COVID awareness in London, broadcasted on BBC. -- <https://www.shapes.ai>.
- Designed and developed UI for mailing system frontend.
- Integrated image recognition API into the mailing system interface.
- Developed E2E CI/CD workflow using bitbucket pipelines with Cypress for automation testing per container.
- Created D3JS dashboard that warns COVID risks in train stops for Indonesian Railway Company (KCI).

PT. Inti Sinergi Teknologi, Jakarta - Software Company

February – June 2019, May – July 2020

Research Assistant / Software Developer

- Developed an end-to-end machine-learning app using Python, Tensorflow, and Keras, converting physical photo IDs into digital text.
- Single-handedly built a self-developed CRNN neural network with an accuracy of 95% that is the central piece of the photo ID app.
- Created the web interface for the machine learning model in Flask / HTML enabling users to use the ML tool in real-time.
- Submitted a white paper with a team of three, containing technical findings of the neural network project.
- Built machine learning model that converts images of electricity meters (PLN) into digital form with 80% accuracy using 1 million+ self annotated image data.
- Built backend server REST API to serve the model above and company projects using MongoDB, Werkzeug, Bcrypt, and FlaskJWT.
- Conducted user load testing on docker swarm using locust and multiple docker containers with 6 nodes (1 manager, 5 workers).

Seriotec GmbH, München, Germany - Media/Film Company

March – August 2018

Full-Stack Developer Intern

- Built and optimized the front-end features for the company website using HTML/CSS, PHP, JS with Processwire -- <https://Yamdu.com>.
- Developed live news feed feature for the company, displaying recent news and other web-app information -- <https://Deinkinoticket.de>.
- Created and trained a chatbot in Facebook Messenger that allows customers to purchase cinema tickets via Facebook.

PT. Indocement Tunggal Prakarsa. Tbk, Jakarta - Cement Company (IT Division)

November 2016 – February 2017

Web Developer Intern

- Introduced and created ASP.NET C# as front-end replacement for better dashboard performance and real-time data.
- Improved web supervision of 50+ cement factories by providing real-time data using SignalR.
- Single handedly created a new chat feature for improved communications via dashboard between factories.

EDUCATION

M. Software Engineering: The University of Melbourne, Melbourne, Australia

December 2024

B.S. Information Technology: Swiss German University, Alam Sutera, Indonesia

August 2019

B.Eng. Engineering Management: Fachhochschule Südwestfalen, Iserlohn, Germany (*Dual Degree*)

August 2019

- Subjects: Artificial Intelligence, Software Engineering, Databases, OOP, Web Systems and Technology
- Publication: Purnama, James, Galinium, Maulahikmah, **Cornelius, Eunico**. (2019). *Development of Convolutional Recurrent Neural Network (CRNN) deep learning model for Indonesian optical character recognition back end web service*. Swiss German University.

PROJECTS

Personal Projects [Portfolio \(eunico.org\)](https://eunico.org)

April 2020 – Current

- Created React Native game (Expo) and created a developers forum using MERN stack development (MongoDB, Express, ReactJS, NodeJS).

VR Exhibition

October 2017

- Developed in Unity and C# a VR showroom app in a team of three, enabling users to build, customize, and view their dream car.

Game-a-thon – Organizer and Designer of a Programming Contest

September 2016

- Organized a video game programming competition that was attended by 600+ high-schoolers from over 22 schools.
- Led game programming workshops for 100+ students using *Gamesalad* -- a game development tool for beginners.